

Compatibility check successful (0)

```
sukesh273@LAPTOP-PI08CKLL:~$ mkdir -p ~/OS-Fourth-Program
```

```
sukesh273@LAPTOP-PI08CKLL:~$ pwd
```

```
/home/sukesh273
```

```
sukesh273@LAPTOP-PI08CKLL:~$ cd ~/OS-Fourth-Program
```

```
sukesh273@LAPTOP-PI08CKLL:~/OS-Fourth-Program$ pwd
```

```
/home/sukesh273/OS-Fourth-Program
```

```
sukesh273@LAPTOP-PI08CKLL:~/OS-Fourth-Program$ code .
```

```
sukesh273@LAPTOP-PI08CKLL:~/OS-Fourth-Program$ |
```

program4.py X

program4.py

```
1  from threading import Thread, current_thread
2  from multiprocessing import Process, Pipe, Value
3
4  processes = [
5      {"pid": "P1", "arrival": 0, "burst": 7},
6      {"pid": "P2", "arrival": 2, "burst": 4},
7      {"pid": "P3", "arrival": 4, "burst": 1},
8      {"pid": "P4", "arrival": 5, "burst": 4},
9  ]
10
11 intervals = [
12     ("P1", 0, 2), ("P2", 2, 4), ("P1", 4, 6),
13     ("P3", 6, 7), ("P2", 7, 9), ("P4", 9, 11),
14     ("P1", 11, 13), ("P4", 13, 15), ("P1", 15, 16)
15 ]
16
17
18 def calculate_metrics():
19     print("\n--- Scheduling Metrics ---")
20     print("PID AT BT CT TAT WT RT")
21
22     tat_total = wt_total = rt_total = 0
23
24     for p in processes:
25         runs = [x for x in intervals if x[0] == p["pid"]]
26         first_start = runs[0][1]
27         completion = runs[-1][2]
28
29         tat = completion - p["arrival"]
```

SL: Ubuntu-24.04 0 0 0

program4.py X

program4.py

```
18 def calculate_metrics():
24     for p in processes:
29         tat = completion - p["arrival"]
30         wt = tat - p["burst"]
31         rt = first_start - p["arrival"]
32
33         tat_total += tat
34         wt_total += wt
35         rt_total += rt
36
37         print(p["pid"], p["arrival"], p["burst"],
38             completion, tat, wt, rt)
39
40     n = len(processes)
41
42     print("Average TAT:", round(tat_total / n, 2))
43     print("Average WT:", round(wt_total / n, 2))
44     print("Average RT:", round(rt_total / n, 2))
45
46
47 def show_gantt_chart():
48     print("\n--- ROUND ROBIN GANTT CHART ---")
49     print(" | ".join(pid for pid, _, _ in intervals))
50     times = [intervals[0][1]] + [end for _, _, end in intervals]
51     print(" ".join(str(t) for t in times))
52
53
54 def thread_task(name):
55     print(name, "running | Thread ID:", current_thread().ident)
```

L: Ubuntu-24.04 (X) 0 / 0 0

program4.py X

program4.py

```
56
57
58 def thread_demo():
59     print("\n--- THREAD DEMO ---")
60     t1 = Thread(target=thread_task, args=("Thread 1",))
61     t2 = Thread(target=thread_task, args=("Thread 2",))
62
63     t1.start()
64     t2.start()
65
66     t1.join()
67     t2.join()
68
69     print("Both threads completed.")
70
71
72 def child_pipe(conn):
73     conn.send("Hello Parent - message from Child")
74     conn.close()
75
76
77 def pipe_demo():
78     print("\n--- PIPE IPC DEMO ---")
79     parent_conn, child_conn = Pipe()
80
81     child = Process(target=child_pipe, args=(child_conn,))
82     child.start()
83
84     message = parent_conn.recv()
```

: Ubuntu-24.04 (X) 0 (A) 0 (M) 0

program4.py X

program4.py

```
77 def pipe_demo():
84     message = parent_conn.recv()
85
86     child.join()
87
88     print("Parent received:", message)
89
90
91 def update_shared(value):
92     value.value += 10
93
94
95 def shared_memory_demo():
96     print("\n--- SHARED MEMORY DEMO ---")
97
98     shared_value = Value("i", 5)
99
100    print("Before child process:", shared_value.value)
101
102    child = Process(target=update_shared, args=(shared_value,))
103    child.start()
104    child.join()
105
106    print("After child process :", shared_value.value)
107
108
```

```
108
109  if __name__ == "__main__":
110      calculate_metrics()
111      show_gantt_chart()
112      thread_demo()
113      pipe_demo()
114      shared_memory_demo()
```

```
● suresh273@LAPTOP-PI08CKLL:~/OS-Fourth-Program$ pwd
/home/suresh273/OS-Fourth-Program
● suresh273@LAPTOP-PI08CKLL:~/OS-Fourth-Program$ python3 program4.py

--- Scheduling Metrics ---
PID AT BT CT TAT WT RT
P1 0 7 16 16 9 0
P2 2 4 9 7 3 0
P3 4 1 7 3 2 2
P4 5 4 15 10 6 4
Average TAT: 9.0
Average WT: 5.0
Average RT: 1.5

--- ROUND ROBIN GANTT CHART ---
P1 | P2 | P1 | P3 | P2 | P4 | P1 | P4 | P1
0 2 4 6 7 9 11 13 15 16

--- THREAD DEMO ---
Thread 1 running | Thread ID: 137833763305152
Thread 2 running | Thread ID: 137833754912448
Both threads completed.

--- PIPE IPC DEMO ---
Parent received: Hello Parent - message from Child

--- SHARED MEMORY DEMO ---
Before child process: 5
After child process : 15
● suresh273@LAPTOP-PI08CKLL:~/OS-Fourth-Program$
```